As large language models (LLMs) advance, scaling data in both quality and quantity becomes crucial for developing generalist models and superintelligent AI. While LLMs have absorbed vast amounts of knowledge from nearly all web data, I believe true intelligence lies in **scientific innovation** conducted within the bounds of **human ethics**. This combination uncovers new knowledge while ensuring it benefits humanity. Despite their impressive performance across diverse tasks, current LLMs still fall far short of human capabilities in expert-level domains such as scientific discovery, Olympic games [1], and real-world engineering. Bridging this gap requires significant progress in **data-centric research**, as data embodies all human wisdom and serves as the foundation of intelligence. Scaling laws based on data quantity and model parameters have nearly reached their limits, yet those based on data quality offer ample room for exploration. This motivates my long-term ambition to explore **scalable high-quality data engineering** and **responsible data usage**. When AI can assist in scientific innovation—including its own research—a powerful positive feedback loop is established. Scalable high-quality data engineering supplies the high-quality data needed for this cycle, while responsible data usage ensures its health and sustainability.

**Future Plans** I envision data-centric research to developing superintelligent AI that emphasizes three key strategies: 1) conducting **scalable pre-training data refinement** to extract more knowledge even when web data runs out; 2) acquiring human cognitive experience through **cognition engineering applied to post-training data**; and 3) ensuring **responsible data usage**.

**Scalable Pre-training Data Refinement**

**Background:** Pre-training is crucial for LLMs to acquire knowledge. As web data becomes depleted, scaling laws based on data quantity are reaching their limits. However, enhancing data quality presents significant potential. High-quality data enables models to gain more knowledge and intelligence without increasing data volume or model size. Traditionally, improving data quality has relied heavily on human expertise and manual adjustments, such as heuristic-based pipelines for document filtering and perplexity scoring. These approaches are labor-intensive and limited by their narrow rule coverage. Alternatively, some research leverages LLMs themselves for high-quality data acquisition through data selection and data synthesis. Yet, data selection reduces the amount of training data, quickly exhausting limited datasets. Data synthesis inherits issues like hallucination from LLMs and faces scalability challenges due to substantial computational demands. These challenges motivate me to enhance pre-training data quality through refinement using smaller LLMs. Inspired by my previous work on post-training data refinement [2], which significantly improved performance across various LLM capacities, I aim to apply similar techniques to pre-training data.

**Proposal:** I propose that pre-training functions as data storage, and the way data is stored determines the efficiency of knowledge acquisition. Cognitive science suggests that well-structured text significantly enhances the human ability to extract key information and comprehend content. This is demonstrated in my paper *Reformatted Alignment* [2], where simply reformatting the response structure improved LLaMA-2-13B's mathematical reasoning accuracy on GSM8K from 46.77% to 56.63%. Building on this, I will explore restructuring pre-training data from two perspectives: first, by using a smaller LLM to reorganize data based on human-preferred structures; second, by applying a retrieval technique from my previous work [3] to pack documents based on semantic similarity, thereby enhancing understanding and reasoning over their full contexts [4]. However, processing billions of pre-training documents with LLMs poses challenges in efficiency and scalability. Addressing this is one of my subgoals, leveraging my experience in efficient NLP [5, 6] and strong-to-weak distillation [7].

**Cognition Engineering of Post-training Data**

**Background:** Human cognition employs System 2 thinking to address complex tasks, encompassing both internal and external processes. Internally, we handle complex problems through mental trial and error—reflection, backtracking, and correction. Externally, we tackle tasks by interacting with the environment and communicating with others. Through both internal and external feedback, humans learn, reflect, and adjust. However, current LLMs cannot reflect and correct their thoughts during infer-

ence time, indicating a lack of System 2 thinking. While the Chain of Thought (CoT) approach provides intermediate steps, it lacks the ability to reflect, engage in trial and error, or learn from feedback. I believe this limitation stems from the post-training data representing a form of shortcut learning, containing only one correct path without trial-and-error steps. If we imagine problem-solving as a search tree, the current data preserves only the successful path, whereas the ideal data should encompass the entire search process. Humans often learn from failures encountered during the search.

**Proposal:** I propose to incorporate human cognitive processes into post-training data through a method called cognition engineering. To simulate external cognition, I will use external tools to obtain cognitive information—specifically agents, feedback mechanisms, and fusion methods. The agents—humans, specialized LLMs, and executors—will generate feedback. Feedback will be derived from data labels, human-AI collaborative outputs, model-based judgments [7], and executor results. This feedback will be integrated using reranking, sequential, or tree-like structures. In my previous work [2], I significantly improved mathematical reasoning by incorporating intermediate thought steps, highlighting the importance of human cognitive processes. For internal cognition, I aim to leverage the thinking processes elicited in LLMs through System 2-style prompting to obtain cognitive information. Additionally, I plan to employ reinforcement learning to enable the model to explore complex tasks, self-evaluate, and ultimately evolve to achieve human-like cognitive capabilities.

**Responsible Data Usage**

**Background:** LLMs face widespread legal scrutiny, primarily because training on copyrighted content or private information may infringe on data producers' rights. At the core of this issue is the inherent trade-off between legal risk and model performance. Given that much of this copyrighted data is created by human experts and is of high quality, not using it for training could significantly degrade model performance [8]. To achieve higher performance, some individuals deliberately manipulate these data and even benchmark test sets to evade detection. Despite significant efforts to detect improper data usage [9], these methods can be easily evaded through iterative rephrasing, translation, or altering the order [10]. For example, while our approach to detecting benchmark data leakage can identify rephrased contamination [11], it remains vulnerable to being circumvented through iterative rephrasing. This challenge motivates me to develop a more robust data detection framework.

**Proposal:** I believe the core issue of data contamination lies in shortcut learning, where models acquire non-generalizable knowledge by leveraging superficial patterns in data, including original or transformed forms (e.g., translations, paraphrases, summaries). For example, contaminated models will become confused when all answer choices appear correct if incorrect options are replaced with correct ones sourced from other questions. This highlights the need for a detection method centered on generalizable knowledge. By distinguishing between generalizable and non-generalizable knowledge, we could significantly simplify the detection of data contamination.

**Motivation for Pursuing a PhD**

My previous research on data-centric AI has reinforced my vision and the research goals I've outlined motivate me to pursue a Ph.D. Long-term, I aim to become a professor dedicated to impactful research and mentorship. Reflecting on my journey, I am inspired by the guidance of my advisors, which has shaped my broader research perspective—a legacy I hope to continue for future scholars.

**My Fitness to University of Massachusetts Amherst**

UMass's top AI program and the vibrant community of students and faculty offer the ideal environment for my research aspirations. Specifically, I am interested in working with 1) **Prof. Andrew McCallum**, 2) **Prof. Hamed Zamani**, 3) **Prof. Hong Yu**, and 4) **Prof. Brendan T. O'Connor** to explore scalable high-quality data engineering and responsible data usage for LLMs, VLMs, and their applications, as I believe high-quality data-centric research is crucial in next-generation AI. I believe my **in-depth research experience**, **solid engineering and research skills**, **willingness to develop real and cutting-edge technology**, and **collaborative mindset** make me a clear fit for UMass.

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